Scheming Schemes

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1 Introduction

This talk is based off Chapter 3 of *The Rising Sea* [Vakil, 2025, p. 65-88]. The goal is to not necessarily introduce schemes in complete generality, but rather with enough information to understand their utility.

In general, we will define a scheme to be the following data:

1. The set X of points on the scheme;
2. The topology / open sets of the scheme;
3. The structure sheaf $\mathcal{O}_X(\cdot)$ of "algebraic functions" on the scheme.

The following will focus on the first two points. Section 2 will be dedicated to understanding the spectrum of a ring, the underlying set of an affine scheme. Section 3 will close the talk with some remarks on the underlying topology of an affine scheme.

For a more in-depth historical motivation for schemes, one may consider [McLarty, 2016].

2 Understanding $\mathrm{Spec}(\cdot)$

2.1 Definitions and Norms

Let A be a commutative ring with "1".

Definition 2.1 (The spectrum)**.** *We denote by* $\mathrm{Spec}(A)$ *the set of prime ideals of the ring* A*.*

Remark 2.2. *As of now,* $\mathrm{Spec}(A)$ *is defined as a set. Later on it will be endowed with a topology and a sheaf of rings will be defined on it.*

Terminology (Ring Elements as Functions): Elements $a \in A$ will be called *functions on* $\mathrm{Spec}(A)$, where their value at $\mathfrak{p} \in \mathrm{Spec}(A)$ will be

$$a(\mathfrak{p}) := a \mod \mathfrak{p}.$$

This gives us the following "translation":

$$\begin{aligned} &a \in A \text{ lying in } \mathfrak{p} \in \mathrm{Spec}(A) \\ &\qquad \leftrightarrow a \in A \text{ a function vanishing at } \mathfrak{p} \in \mathrm{Spec}(A) \end{aligned}$$

Exercise 2.3. *For $a, b \in A$, show that $(a+b)(\mathfrak{p}) = a(\mathfrak{p}) + b(\mathfrak{p})$ and $(ab)(\mathfrak{p}) = a(\mathfrak{p})b(\mathfrak{p})$.*

Sol. →

Terminology (Coordinates): If A is generated over a base ring by $x_1, \ldots, x_r$, then these are often called the coordinates of A.

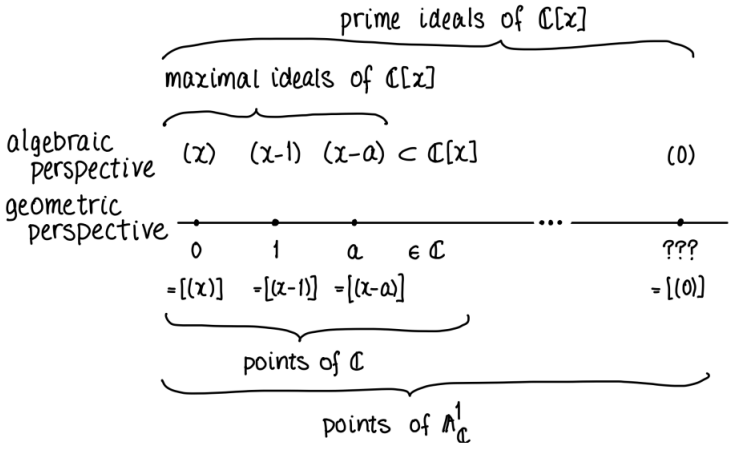
2.2 Some examples

We now consider a couple of examples.

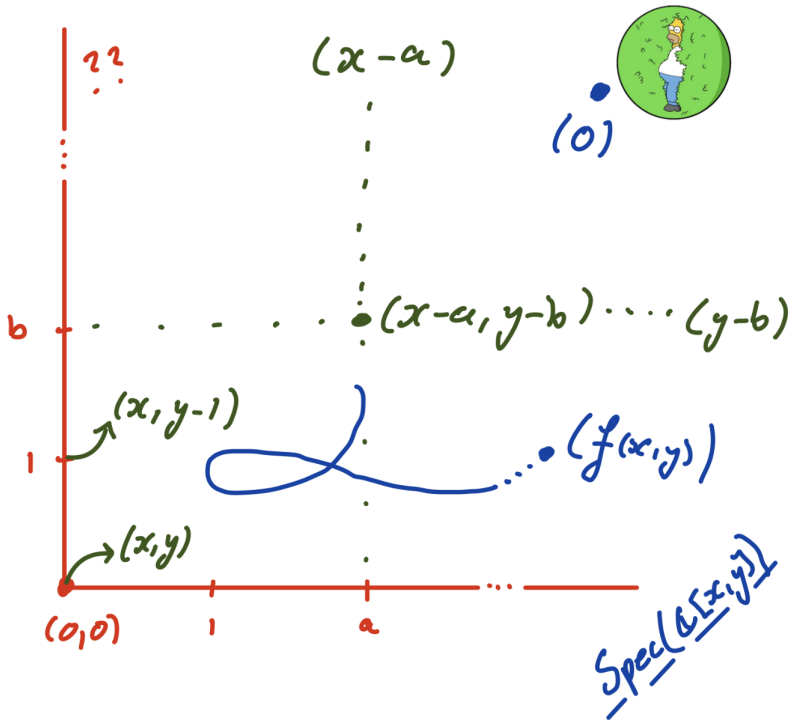
Example 2.4 (The Complex Line $\mathbb{A}^1_{\mathbb{C}}$)**.** *We wish to apply the above translation to this example.*

- *What are the most natural functions on $\mathbb{C}$? Polynomials $P \in \mathbb{C}[x]$.*
- *For any $a \in \mathbb{C}$, which $P \in \mathbb{C}[x]$ vanish at a? Those which are in the prime ideal $\mathfrak{p}_a = (x-a) \subset \mathbb{C}[x]$.*

Thus, using our translation, the correspondence is clear between $\mathbb{A}^1_{\mathbb{C}} = \mathrm{Spec}(\mathbb{C}[x])$. We further visualise this translation as such [Vakil, 2025, p. 68]:



Example 2.5 (The (Complex) Complex Plane)**.** *We use the above example to get a similar picture of* $\mathrm{Spec}(\mathbb{C}[x,y])$*. Note the generic point* (0) *again in the top right corner.*



A few things to note before continuing.

- One traditional point / prime for each $a \in \mathbb{C}$ (or $(a,b) \in \mathbb{C}^2$).
- Both instances, we got an additional point (0) – "generic point" (to be defined later).
- How to think of (0): somewhere on the complex line, but nowhere in particular.
- Later we will see that it is indeed "close" to all the other points.
- We also have the primes in $\mathrm{Spec}(\mathbb{C}[x,y])$ that are not maximal. For example, the vertical line corresponding to $(x-a)$, the horizontal line corresponding to $(y-b)$, and the (irreducible) equation $(f(x,y))$ (e.g. $f(x,y) = y - x^2$ is irreducible).
- Note as well: maximal ideals correspond to small points. Broadly, if $\mathfrak{p}, \mathfrak{p}_0 \in \mathrm{Spec}(\mathbb{C}[x,y])$, then $\mathfrak{p}_0 \subseteq \mathfrak{p}$ corresponds to p_0 being a "bigger" point.

The above description of $\mathbb{A}^2_{\mathbb{C}} = \mathrm{Spec}(\mathbb{C}[x,y])$ is also an exhaustive classification of the points.

Exercise 2.6. *Show that we did not "miss" any primes when discussing* $\mathbb{A}^2_{\mathbb{C}} = \operatorname{Spec}(\mathbb{C}[x,y])$ *in the following way:*

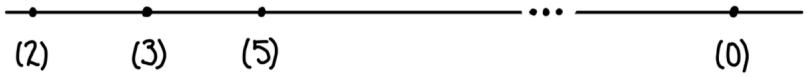
a. *Suppose there is a non-principal prime* $\mathfrak{p} \in \operatorname{Spec}(\mathbb{C}[x,y])$ *that we did not list above. Use the fact that* $\mathbb{C}[x,y]$ *is a UFD and the primality of* $\mathfrak{p}$ *to show that there exist coprime elements* $f, g \in \mathfrak{p}$.

b. *Use the Euclidean algorithm in the Euclidean domain* $\mathbb{C}(x)[y]$ *to show that you can find a nonzero* $h \in (f,g) \subset \mathfrak{p}$.

c. *Finally, use the primality of* $\mathfrak{p} \cap \mathbb{C}[x]$ *to show that* $(x-\alpha)$ *is in* $\mathfrak{p}$ *for some* $\alpha \in \mathbb{C}$. *Use a similar argument to show that* $(y-\beta)$ *is in* $\mathfrak{p}$ *for some* $\beta \in \mathbb{C}$. *Then, conclude that* $\mathfrak{p}$ *was already on the list.*

Sol. →

Example 2.7 ($\operatorname{Spec}(\mathbb{Z})$)**.** *This space will look similar to* $\mathbb{C}[x]$ *(or, more generally* $K[x]$ *where* $K = \overline{K}$*). Both* $\mathbb{Z}$ *and* $K[x]$ *are UFDs, and the primes are* (0) *(again, the general point) and* (p) *for* $p \in \mathbb{Z}$ *a prime. Thus, we essentially have the same picture as for* $\operatorname{Spec}(\mathbb{C}[x])$*, but with each* $p \in \mathbb{Z}$ *and* 0 *corresponding to each point [Vakil, 2025, p. 69]:*



Functions: What are some of the functions on these spaces? Recall that for $a \in A$ and $\mathfrak{p} \in \operatorname{Spec}(A)$, $a(\mathfrak{p}) := a \mod \mathfrak{p}$.

Example 2.8. *Take any* $f \in \mathbb{C}[x]$. *Then,* $f(\mathfrak{p}) := f \mod \mathfrak{p}$.

- *If* $\mathfrak{p} = (x-\alpha)$*, then this merely corresponds to evaluating* f *at* $x = \alpha$.
- *If* $\mathfrak{p} = (0)$*, then* $f((0)) = f(x)$.
- *If we instead take some* $f \in \mathit{Frac}(\mathbb{C}[x])$*, say* $f = (x-3)^3/(x-2)$*, we will eventually recognise this as being a function of the structure sheaf on the open set* $\mathbb{A}^1_{\mathbb{C}} \setminus \{2\}$.

Example 2.9. *Take any $f \in \mathbb{C}[x,y]$. A lot of the previous example still applies here. We consider a new case.*

- *Take $f = y^2 - x^4$ and $\mathfrak{p} = (y - x^2)$. Then:*

$$\begin{aligned} f(\mathfrak{p}) &= y^2 - x^4 \mod (y - x^2) \\ &= (y - x^2)(y + x^2) \mod (y - x^2) \\ &= 0 \mod (y - x^2) \end{aligned}$$

Example 2.10. *Consider $a \in \mathbb{Z}$, then a is a function on $\mathrm{Spec}(\mathbb{Z})$ where $a(\mathfrak{p}) \equiv a \bmod \mathfrak{p}$. For example:*

$$10((3)) \equiv 1 \mod 3.$$

Similarly, we have rational functions, such as $27/4$ defined away from $(2) \in \mathrm{Spec}(\mathbb{Z})$ with a "triple root" at (3). Then:

$$\frac{27}{4}((5)) \equiv 27(4^{-1}) \mod 5 \equiv 3 \mod 5.$$

A couple more examples will be left as "exercises" to ponder on.

Exercise 2.11. *Describe the points of $\mathbb{A}^1_{\mathbb{R}} := \mathrm{Spec}(\mathbb{R}[x])$. For $\mathfrak{p} \in \mathrm{Spec}(\mathbb{R}[x])$, give some examples of $f(\mathfrak{p})$ for $f \in \mathbb{R}[x]$.*

Sol. →

Exercise 2.12. *Describe the sets $\mathbb{A}^1_{\mathbb{Q}} = \mathrm{Spec}(\mathbb{Q}[x])$ $\mathbb{A}^1_{\mathbb{F}_q} = \mathrm{Spec}(\mathbb{F}_q[x])$. Deduce which primes correspond to the elements of $\mathbb{Q}$ and $\mathbb{F}_q$ respectively, and what the other points correspond to.*

Sol. →

Exercise 2.13. *Adapt Euclid's proof of the infinitude of the primes to show that $\mathrm{Spec}(K[x])$ has infinitely many points (Hint: recall that $K[x]$ is a unique factorisation domain).*

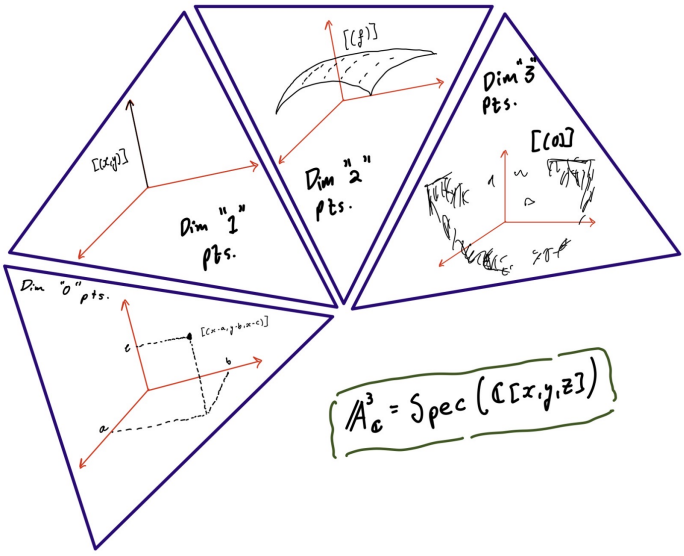
Sol. →

2.3 Generalising these examples

Definition 2.14. *For A an arbitrary ring, we write $\mathbb{A}^n_A$ for the (affine) scheme* $\mathrm{Spec}(A[x_1, \ldots, x_n])$.

Naturally, we may want to imagine how the schemes $\mathbb{A}^n_A$ "look". In a last chance to get some visual intuition, we set $A = \mathbb{C}$ and $n = 3$.

Example 2.15. *In terms of three-dimensional points we have the generic point* $(0) \in \mathrm{Spec}(\mathbb{C}[x, y, z])$. *Two-dimensional points correspond to "hypersurfaces" (f) ($f \in \mathbb{C}[x.y, z]$ irreducible) and zero-dimensional points to our typical coordinates* $(x - \alpha, x - \beta, x - \gamma) \in \mathrm{Spec}(\mathbb{C}[x, y, z])$ *($\alpha, \beta, \gamma \in \mathbb{C}$). Without invoking any machinery too heavy for where we are now, one dimensional points here will correspond to what we think of as irreducible curves (thus turning an algebraic question into a geometric one).*



Example 2.16. *We generalise a previous exercise on interpreting the points of $\mathbb{A}^1_{\mathbb{Q}}$ corresponding to maximal ideals as points of $\overline{\mathbb{Q}}$ whose Galois conjugates are glued together for $\mathbb{A}^n_{\mathbb{Q}}$ (for $n = 2$). We intend to illuminate this by showing that $(\sqrt{2}, \sqrt{2})$ $\leftarrow$ is glued to $\rightarrow$ $(-\sqrt{2}, -\sqrt{2})$, but not to, say, $(\sqrt{2}, -\sqrt{2})$.*

The former points both correspond to $\mathfrak{p} = (x^2 - 2, y - x)$, whilst the other point (and its negative / Galois conjugate) corresponds to $\mathfrak{p}' = (x^2 - 2, y + x)$. However, both have residue fields $\mathbb{Q}[x, y]/(x^2 - 2, y - x) \cong \mathbb{Q}(\sqrt{2}) \cong \mathbb{Q}[x, y]/(x^2 - 2, y + x)$.

Intuitively, what this tells us is that the ideal $(x^2 - 2, y + x) \in \operatorname{Spec}(\mathbb{Q}[x, y]) = \mathbb{A}^2_{\mathbb{Q}}$ corresponds to two "classical" geometric points "glued" together by a Galois action. One way to think about this is, in the case of $\mathbb{A}^2_{\mathbb{C}}$, the traditional geometric points $(\sqrt{2}, \sqrt{2})$ and $(-\sqrt{2}, -\sqrt{2})$ correspond to distinct primes $(x \pm \sqrt{2}, y \pm \sqrt{2})$, whilst in $\mathbb{A}^2_{\mathbb{Q}}$ the prime $(x^2 - 2, y - x)$ corresponds to this "fuzzy" point consisting of $(\sqrt{2}, \sqrt{2})$ and its Galois conjugates.

2.4 Spec(·), quotients, and localisation

We recall the fourth / lattice isomorphism theorem.

Theorem 2.17. *Let $\mathfrak{a}$ be an ideal of the ring A. There is an inclusion preserving bijection between*

$$\mathscr{A} = \{\mathfrak{b} \in Idl(A) : \mathfrak{a} \subseteq \mathfrak{b}\}$$

and

$$\overline{\mathscr{A}} = \{\overline{\mathfrak{b}} \in Idl(A/\mathfrak{a})\}.$$

Proof. (Sketch of a proof) Use the projection map $\pi : A \to A/\mathfrak{a}$ with $\pi(\mathfrak{b}) = \mathfrak{b}/\mathfrak{a} =: \overline{\mathfrak{b}} \in \overline{\mathscr{A}}$ (for $\mathfrak{b} \in \mathscr{A}$). Note that $\pi^{-1}(\overline{\mathfrak{b}}) \in \operatorname{Idl}(A)$ and $\ker \pi = \mathfrak{a} \subseteq \pi^{-1}(\overline{\mathfrak{b}}) := \mathfrak{b} \in \mathscr{A}$. □

Replacing "ideals" (or subrings) with "prime ideals" tells us that $\operatorname{Spec}(A/\mathfrak{a}) \subseteq \operatorname{Spec}(A)$. The following exercise also demonstrates some of the interplay between "Spec" and ring homomorphisms.

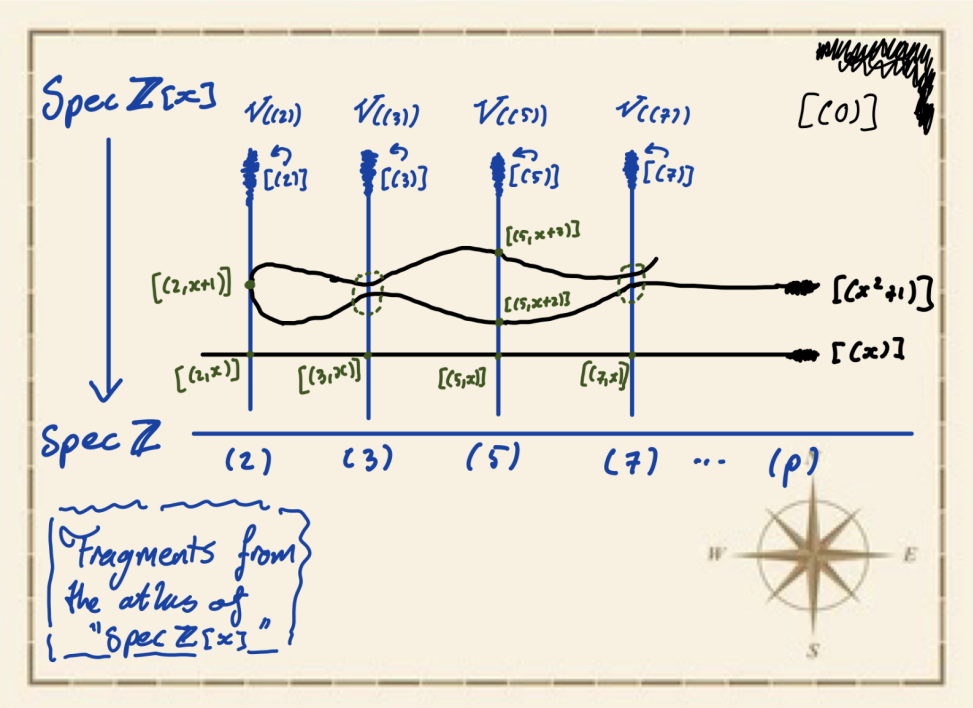
Exercise 2.18. *For $\varphi : A \to B$ a ring homomorphism, show that the natural structural morphism $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$ induces the map of sets $\operatorname{Spec}(\varphi) : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$.*

Sol. →

We illuminate these new techniques with the following pictures / examples, the first concerning itself with the "atlas of Spec $\mathbb{Z}[x]$" [Belmans, 2011] and the Gaussian integers.

Example 2.19. *We already saw that* $\operatorname{Spec}\mathbb{Z}$ *"looked" similar to* $\operatorname{Spec}\mathbb{C}[x]$*, so how can we picture* $\operatorname{Spec}(\mathbb{Z}[x])$*? Instead we view* $\operatorname{Spec}(\mathbb{Z}[x])$ *as a scheme "over"* $\operatorname{Spec}(\mathbb{Z})$ *via* $\operatorname{Spec}\varphi : \operatorname{Spec}(\mathbb{Z}[x]) \to \operatorname{Spec}(\mathbb{Z})$*. There are a couple of things to note:*

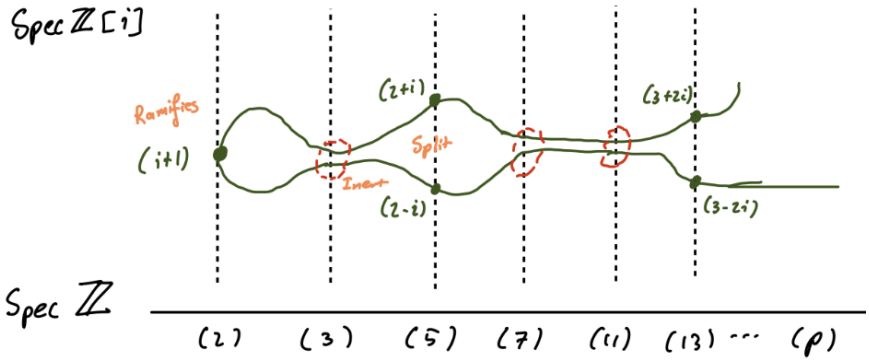
- *The generic point* (0) *is again depicted with a "fuzziness". The other points* $\mathfrak{p}$ *also have this quality as they "tend to infinity", signifying being their own generic point on the scheme* $\operatorname{Spec}\mathbb{Z}[x]/\mathfrak{p}$*.*
- *We have some irreducible curves (e.g.* (x^2+1)*,* (x)*). Whilst looking different to their regular* $\mathbb{R}^2$ *depiction, here there is a greater emphasis on how they interact with the points* $(p) \in \operatorname{Spec}\mathbb{Z}[x]$*.*
- *The vertical lines correspond to* $(p) \in \operatorname{Spec}\mathbb{Z}[x]$ *(*$p \in \mathbb{Z}$*). What is more interesting is their "intersection" with the irreducible curves corresponding to the maximal ideals* $\mathfrak{m} = (p, f(x))$*. For their intersection with* $f(x) = x$ *we always have* (p, x) *being maximal. However, for* $f(x) = x^2 + 1$ *this is not guaranteed. For example,* $\mathbb{Z}[x]/(2, x^2+1) \cong \mathbb{F}_2[x]/(x^2+1) \cong \mathbb{F}_2[x]/(x+1)^2$ *(as we compute everything modulo 2). Thus, the maximal here corresponds to* $\mathfrak{m} = (2, x+1)$*. This is also the case with other primes in the depiction.*



Example 2.20. *With the previous example in mind, let's consider* $\operatorname{Spec}\mathbb{Z}[i]$ *by "zooming" into* $\operatorname{Spec}\mathbb{Z}[x]$ *via taking the quotient* $\mathbb{Z}[x]/(x^2+1) \cong \mathbb{Z}[i]$*. Doing so may also serve to illuminate the "geometry" behind the behaviour of the primes in the extension* $\mathbb{Q}(i) \mid \mathbb{Q}$*.*

The primes which ramify are those which have a "double root" with x^2+1*. In this case,* $(2)\mathbb{Z} = (1+i)^2\mathbb{Z}[i]$*. The primes which are inert are those which intersect the curve once (note the big circle hopefully conveying that* (3) *intersects only at one point). Then, for example,* $(3)\mathbb{Z} = (3)\mathbb{Z}[i]$*. Primes which are* $1 \mod 4$ *split into a product of two primes in* $\mathbb{Z}[i]$*, such as* $5\mathbb{Z} = (2+i)(2-i)\mathbb{Z}[i]$*, depicted by intersecting the curve more than once. We will come back to this example at the end of this section.*





Remark 2.21. *Whilst the previous example explained "taking a quotient" as "zooming in", we are explicitly zooming in on (x^2+1) and chiselling away until we are just left with said curve. Even the vertical lines $V((p))$ from the previous example now just reduce to their intersections with (x^2+1). In later examples involving localisation, we will also use the "zooming in" metaphor, but in a more typical "looking through a microscope" way.*

Exercise 2.22. *The following exercise attempts to illuminate and formalise the translation between maps given by explicit formulas and maps of rings.*

a. *Let (a, b) be the coordinates for $\mathbb{C}^2$ and (x, y, z) for $\mathbb{C}^3$. If we wanted to consider a map from the standard parabola in $\mathbb{C}^2$ to the curve in $\mathbb{C}^3$ cut out by $y = x^2$ and $z = y^2$, we have a map of schemes*

$$\operatorname{Spec} \mathbb{C}[a, b]/I \to \operatorname{Spec} \mathbb{C}[x, y, z]/J.$$

What are I and J, and what would the morphism of $\mathbb{C}$-algebras $\mathbb{C}[x, y, z]/J \to \operatorname{Spec} \mathbb{C}[a, b]/I$ look like?

b. *Let $\varphi : A \to B$ be a ring homomorphism and $\mathfrak{a} \in Idl(A)$, $\mathfrak{b} \in Idl(B)$ with $\varphi(\mathfrak{a}) \subseteq \mathfrak{b}$. Show that φ induces a map of sets $\operatorname{Spec}(B/\mathfrak{b}) \to \operatorname{Spec}(A/\mathfrak{a})$ given by $\mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q})$.*

c. *Let $A = K[y_1, \dots, y_n]$ and $B = K[x_1, \dots, x_m]$, with the morphism $\varphi : A \to B$ sending $y_i \mapsto f_i(x_1, \dots, x_m) \in K[x_1, \dots, x_m]$. Show that φ sends the point $(a_1, \dots, a_m) \in K^m$ (or rather, $(x_1 - a_1, \dots, x_m - a_m) \in \operatorname{Spec}(K[x_1, \dots, x_m])$) to $(f_1(a_1, \dots, a_m), \dots, f_n(a_1, \dots, a_m)) \in K^n$.*

Sol. →

We also explore the interplay between $\operatorname{Spec}(\cdot)$ and localisation. Recall that a subset $S \subseteq A$ is multiplicative if $1 \in S$ and $xy \in S$ for all $x, y \in S$. We let $\varphi : A \to S^{-1}A$ be the natural homomorphism given by $\varphi(a) = a/1$.

Exercise 2.23. *Show that the prime ideals of $S^{-1}A$ are in an order preserving one-to-one correspondence ($\mathfrak{p} \overset{\varphi}{\mapsto} S^{-1}\mathfrak{p}$) with the prime ideals of A which don't meet S. You may use the fact that extensions and contractions are order preserving.*

Sol. →

Recall as well the two main forms of localisation:

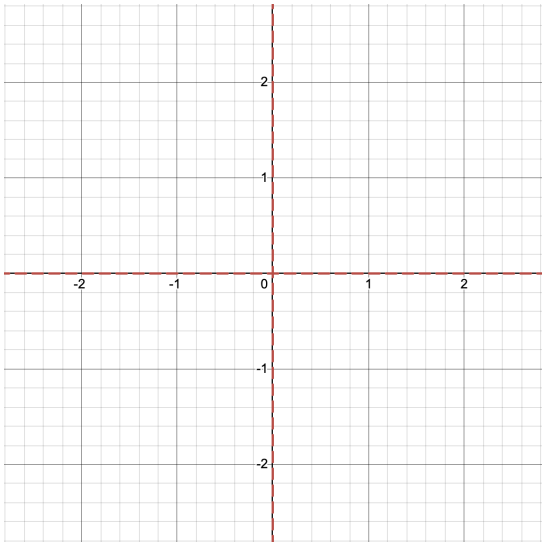
- $A_f = \{1, f, f^2, \dots\}^{-1}A$ where $f \in A$.

- E.g. $A = \mathbb{Z}[i]$ and $f = 3 \in \mathbb{Z}[i]$. Then $\mathbb{Z}[i]_3 = \mathbb{Z}[i, 1/3]$, so powers of 3 are inverted.
- Now, think of the spectrum as the points where 3 does not "vanish". We want $a/3^k \in \mathbb{Z}[i, 1/3]$ to be defined on $\operatorname{Spec}(\mathbb{Z}[i, 1/3])$. As $\frac{a}{3^k}((p)) = 3^{-k}a \mod p$, this is possible for any prime not equal to 3.

- $A_{\mathfrak{p}} = (A \setminus \mathfrak{p})^{-1}A$ where $\mathfrak{p} \in \operatorname{Spec}(A)$.
  - E.g. given $(3) \in \operatorname{Spec}\mathbb{Z}[i]$, we consider $\mathbb{Z}[i]_{(3)}$, a local ring whose unique maximal is (3), and $\operatorname{Spec}(\mathbb{Z}[i]) = \{(0), (3)\}$. In general, the spectrum of $A_{\mathfrak{p}}$ will be the prime ideals of A contained in $\mathfrak{p}$.

Example 2.24. *Let $A = \mathbb{C}[x, y]$ and $f = y - x^2$. Then, $\operatorname{Spec}(\mathbb{C}[x, y]_{y-x^2})$ are the primes not containing f, or rather the primes $\mathfrak{p} \in \operatorname{Spec}(\mathbb{C}[x, y])$ where $f(\mathfrak{p}) \neq 0$ (eventually to be called the "distinguished opens set"). So, $\operatorname{Spec}(\mathbb{C}[x, y]_{y-x^2})$ is $\mathbb{A}^2_{\mathbb{C}}$ with the points on $y - x^2 = 0$ thrown out (corresponding to $(y - a, x - a^2) \in \operatorname{Spec}(\mathbb{C}[x, y])$. The point / curve itself, $(y - x^2) \in \operatorname{Spec}(\mathbb{C}[x, y])$, is also thrown out.*

The next example aims to link localisation with inverting zero divisors.



Example 2.25. *Consider $\operatorname{Spec}(\mathbb{C}[x, y]/(xy))$ (its real points pictured above). Moreover, take $f = x \in \mathbb{C}[x, y]$. Then, as $\operatorname{Spec}(\mathbb{C}[x, y]/(xy))$ is the union of the x and y-axes, localising prompts us to "throw out" the locus where x vanishes. This takes place on the y-axis, so we are left with something like the x-axis without the origin ($\mathbb{A}^1_{\mathbb{C}} \setminus (0)$, which we expect to identify with $\operatorname{Spec}(\mathbb{C}[x]_x)$).*

Exercise 2.26. *Prove that the two rings are isomorphic*

$$(\mathbb{C}[x,y]/(xy))_x \xrightarrow{\sim} \mathbb{C}[x]_x.$$

Hint: consider the projection map

$$\pi : \mathbb{C}[x,y] \to \mathbb{C}[x,y]/(xy), f \mapsto [f],$$

along with $[f(x,y)]/x^n \mapsto f(x,0)/x^n$.

Sol. →

Now, an example with localising at a prime ideal.

Example 2.27. *Let* $A = \mathbb{C}[x,y]$ *and* $\mathfrak{p} = (x,y)$. *Then, the primes of* $A_\mathfrak{p}$ *are, in general, the primes of* A *contained in* $\mathfrak{p}$. *In this case, we are zooming into the origin and focusing on the irreducible curves* $f \in \mathbb{C}[x,y]$ *which pass through the origin* $(f(0,0) = 0)$. *On top of those one-dimensional points* (f), *we also have the zero-dimensional point* (x,y) *and the two-dimensional* (0).

Example 2.28. *Take* $A = \mathbb{Z}$ *and consider the primes* $(0), (2) \in \operatorname{Spec}(\mathbb{Z})$. *Then,* $\operatorname{Spec}(\mathbb{Z}_{(0)})$ *may be seen as zooming into the generic point of* $\operatorname{Spec}(\mathbb{Z})$, *no other points being visible. On the other hand,* $\operatorname{Spec}(\mathbb{Z}_{(2)})$ *has us zooming into the prime* (2) – *the generic point is still visible (we cannot necessarily pinpoint its location anywhere in particular). Thus,* $\operatorname{Spec}(\mathbb{Z}_{(2)}) = \{(0), (2)\}$.

This next fact is somewhat of a surprising one, and may require us to slightly adjust our visual / algebraic intuition of schemes. The fact is that functions are not determined by their values at points.

Exercise 2.29. *Consider the "dual numbers"* $K[\varepsilon]/(\varepsilon^2)$. *Show that* $\operatorname{Spec} K[\varepsilon]/(\varepsilon^2) = \{(\varepsilon)\}$. *Hint: consider* $\operatorname{Spec}(K[x])$ *and use the fact that the primes in* $\operatorname{Spec} K[\varepsilon]/(\varepsilon^2)$ *correspond to those in* $\operatorname{Spec}(K[x])$ *satisfying* $(x^2) \subseteq \mathfrak{p}$.

Sol. →

Example 2.30. *A way to illuminate the above "surprising fact" is with the case of the dual numbers. Indeed,* $\varepsilon([(\varepsilon)]) = \varepsilon \mod \varepsilon \equiv 0$, *but* $\varepsilon \neq 0$.

What is going on here? Consider the two curves $x = 0$ *and* $y = 0$ *meeting at the origin in* $\mathbb{A}^2_K = \operatorname{Spec} K[x, y]$. *Then,* $\operatorname{Spec} K[x, y]/(x, y) = \operatorname{Spec} K$, *which is just one point (the origin). Now, consider the same idea but with the curves* $y = 0$ *and* $y = x^2$. *Here,* $K[x, y]/(y, y - x^2) \cong K[x]/(x^2)$, *where we saw that* $\operatorname{Spec} K[\varepsilon]/(\varepsilon^2) = \{(\varepsilon)\}$. *Now, both spectra only have one point, but the one corresponding to the dual numbers has a nilpotent prime, rendering it "thicker".*

What this means is that the affine scheme of the dual numbers remembers the multiplicity of the intersection – implying that having the same number of points needs not imply that they have the same geometric information.

Example 2.31 (A bit more ANT)**.** *Let* $L \mid K$ *be an extension of number fields where* $L = \mathbb{Q}[x]/(f(x))$. *Given the maps of rings*

$$\mathcal{O}_K \hookrightarrow \mathcal{O}_K[x] \to \mathcal{O}_K[x]/(f(x)) \cong \mathcal{O}_L$$

we get

$$\operatorname{Spec}(\mathcal{O}_L) = \operatorname{Spec}(\mathcal{O}_K[x]/(f(x)) \to \operatorname{Spec}(\mathcal{O}_K[x]) \hookrightarrow \operatorname{Spec}(\mathcal{O}_K).$$

For $\mathfrak{P} \in \operatorname{Spec}(\mathcal{O}_L)$, *the composite map above is then just* $\mathfrak{P} \mapsto \mathfrak{P} \cap \mathcal{O}_K =: \mathfrak{p}$. *Take a prime* $\mathfrak{p} \in \operatorname{Spec}(\mathcal{O}_K)$. *The behaviour of* $\mathfrak{p}$ *in* $\mathcal{O}_L$ *is then encoded by the ring*

$$(\mathcal{O}_K/\mathfrak{p}\mathcal{O}_K)[x]/(f(x)) \cong \mathbb{F}_\mathfrak{p}[x]/(f(x)).$$

Then, the fibre $\pi^{-1}(\mathfrak{p}) = \operatorname{Spec} \mathbb{F}_\mathfrak{p}[x]/(f(x))$ *(where* π *is the above composite map) encodes a lot of the information about* $\mathfrak{p}$ *in the extension. For example, in the case of the Gaussian integers, at* $\mathfrak{p} = (5)$, $\pi^{-1}((5)) = \operatorname{Spec}(\mathbb{F}_5 \times \mathbb{F}_5)$, *whose primes are* $(2 + i) \times \mathbb{F}_5$ *and* $\mathbb{F}_5 \times (2 - i)$. *For* $\mathfrak{p} = (3)$, $\pi^{-1}((3)) = \operatorname{Spec}(\mathbb{F}_9)$ *which is a field.*

At $\mathfrak{p} = (2)$, *we have* $\pi^{-1}((2)) = \operatorname{Spec}(\mathbb{F}_2[x]/(x + 1)^2) = \operatorname{Spec}(\mathbb{F}_2[\varepsilon]/(\varepsilon)^2)$, *corresponding to one "thick" (or "fuzzy") prime above it (i.e. ramification).*

Exercise 2.32. *Let* $K = \mathbb{Q}(\sqrt{d})$ *be a quadratic extension of* $\mathbb{Q}$. *Use the above example to deduce that* $(p) \in \operatorname{Spec} \mathbb{Z}$ *splits (i.e.* $p\mathcal{O}_{\mathbb{Q}(\sqrt{d})} = \mathfrak{p}\overline{\mathfrak{p}}\mathcal{O}_{\mathbb{Q}(\sqrt{d})}$*) in* $\mathcal{O}_{\mathbb{Q}(\sqrt{d})}$ *if and only if* d *is a quadratic residue modulo* p.

Sol. →

Below are some exercises on nilpotents, which are ring elements which have a power that is zero.

Exercise 2.33. *Recall that a ring element r is nilpotent if $r^n = 0$ for some $n > 1$.*

a. *Show that if $\mathfrak{n}$ is an ideal of nilpotents, then $\operatorname{Spec}(A/\mathfrak{n}) \to \operatorname{Spec}(A)$ is a bijection (using the inclusion from a previous exercise).*

b. *Show that the nilpotents of a ring form an ideal $\mathfrak{N} = \mathfrak{N}(A)$, which we call the nilradical of A. Note, $\mathfrak{N}(A)$ is not only contained within the intersection of all the primes, it is equal to it.*

Sol. →

This leads to the following theorem.

Theorem 2.34. *A function on $\operatorname{Spec} A$ vanishes at every point if and only if it is nilpotent.*

3 Topology

We finish this introduction to (affine) schemes by introducing the underlying topological space of an affine scheme. Whilst it may be more common to describe topologies using open sets, here it is more convenient to define the Zariski topology on the spectrum of a ring in terms of closed subsets.

Definition 3.1. *We define the Zariski topology on $\operatorname{Spec}(A)$ in the following way. Let S be a subset of the ring A. Then, we define the vanishing set of S to be*

$$V(S) := \{\mathfrak{p} \in \operatorname{Spec}(A) : S \subset \mathfrak{p}\}.$$

Then, we define the Zariski topology by declaring that $V(S)$ is closed for all S.

Exercise 3.2. *Show that the above definition fits the definition of a topology. In other words, show:*

a. $\varnothing, \operatorname{Spec} A$ *are both open subsets of* $\operatorname{Spec}(A)$.

b. *If* $\mathfrak{a}_i$ *is a collection of ideals, then* $\cap_i V(\mathfrak{a}_i) = V(\sum_i \mathfrak{a}_i)$.

c. $V(\mathfrak{a}_1) \cup V(\mathfrak{a}_2) = V(\mathfrak{a}_1\mathfrak{a}_2)$.

Sol. →

We now recall some properties of the vanishing set function $V(\cdot)$. If any of the following properties are not obvious, they make as good exercises.

- $V(\cdot)$ is inclusion reversing: $S_1 \subset S_2 \implies V(S_2) \subset V(S_1)$.
- $V(\sqrt{\mathfrak{a}}) = V(\mathfrak{a})$.
- As $(\mathfrak{a} \cap \mathfrak{b})^2 \subset \mathfrak{a}\mathfrak{b} \subset \mathfrak{a} \cap \mathfrak{b}$, we have $V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b})$.
- $V(S) = V((S))$ (the vanishing set of S is equal to the vanishing set of the ideal generated by S).
- The *Zariski closure* of a subset $S \subseteq \operatorname{Spec} A$ is the smallest Zariski closed set that contains S. Then, $\overline{S} = \bigcap_{S \subseteq V(\mathfrak{a})} V(\mathfrak{a})$ where the intersection goes over the ideals $\mathfrak{a} \in \operatorname{Idl}(A)$.
- Let $I(S) := \{f \in A : f \in \mathfrak{p} \text{ for every } \mathfrak{p} \in S\}$. Then, $\overline{S} = V(I(S))$.

Example 3.3 (Focusing on generic points)**.** *Consider the generic point* $(0) \in \mathbb{A}^n_K = \operatorname{Spec} K[x_1, \ldots, x_n]$. *Then, its closure is* $\overline{(0)} = V((0)) = \operatorname{Spec} K[x_1, \ldots, x_n]$. *It is in this way that we mean that the generic point is close to every point in* $\mathbb{A}^n_K$*, despite not being able to exactly pinpoint "where" it is.*

Remark 3.4. *It may be useful to also note that each closed irreducible subset has a corresponding generic point sitting on top of it. Intuitively, take* $(0) \in \operatorname{Spec} K[x, y]$. *Then, consider* $V(y - x^2)$. *Taking quotients yields* $K[x, y]/(y - x^2) \cong K[x]$, *where the generic point here is "still"* $(0) \in \operatorname{Spec} K[x]$. *However, zooming out to the whole of* $\operatorname{Spec} K[x, y]$, *its generic point becomes the curve* $(y - x^2)$.

This was made evident in the visualisation of $\operatorname{Spec} \mathbb{Z}[x]$, *where each irreducible polynomial (and, in general, closed irreducible subset) had this "fuzziness" towards the end, corresponding to the generic point sitting on top of it.*

Example 3.5 (The Closed Subsets of $\mathbb{A}^2_{\mathbb{C}}$)**.** *In a previous exercise, we identified all the prime ideals of* $\mathbb{C}[x, y]$*. These were:*

- *The maximal ideals* $(x - a, y - b)$ *("zero-dimensional points");*
- *The one-dimensional points* (f) *(*$f(x, y)$ *irreducible);*
- *The two-dimensional point* (0)*.*

The closed subsets of $\mathbb{A}^2_{\mathbb{C}}$ *are then the entire space (coming from the closure of* (0)*)*

$$Z = (\textit{finitely many irreducible curves}) \cup (\textit{finitely many points}).$$

It could be the case that there is no "irreducible curve" component (respectively "finitely many points" component). The irreducible curves are the closure of a one-dimensional point – the point with the zero-dimensional points laying on it.

We finish with a couple more exercises.

Exercise 3.6. *We saw that* $\varphi : B \to A$ *induces a map of sets* $\pi : \operatorname{Spec}(A) \to \operatorname{Spec}(B)$*. Show that* π *is a continuous map by showing that closed sets pull back to closed sets. Then, interpret* Spec *as a contravariant functor Rings* $\to$ *Top.*

Sol. →

Exercise 3.7. *Let* $\mathfrak{b}, S \subset B$ *be an ideal and multiplicative subset.*

a. *Show that* $\operatorname{Spec}(B/\mathfrak{b})$ *is naturally a closed subset of* $\operatorname{Spec} B$*. If* $S = \{1, f, f^2 \dots\}$*, show that* $\operatorname{Spec}(S^{-1}B)$ *is naturally an open subset of* $\operatorname{Spec}(B)$*.*

b. *Show that for an arbitrary* S*,* $\operatorname{Spec}(S^{-1}B) \subseteq \operatorname{Spec} B$ *need not be open or closed.*

c. *Show that the Zariski topology on* $\operatorname{Spec} B/\mathfrak{b}$ *(resp.* $\operatorname{Spec} S^{-1}B$*) is the subspace topology induced by inclusion in* $\operatorname{Spec}(B)$*.*

Sol. →

Solutions to Exercises

Solution to Exercise 2.2

Use the fact that $\pi_{\mathfrak{p}} : A \to A/\mathfrak{p}$ is a ring homomorphism then show directly.

Solution to Exercise 2.5

(a) Suppose that $\mathfrak{p} \in \operatorname{Spec}(\mathbb{C}[x,y])$ is not principal. Then, there exist coprime elements $f, g \in \mathbb{C}[x,y]$. This is guaranteed as $\mathbb{C}[x,y]$ is a UFD and $\mathfrak{p}$ being prime tells us that $ab \in \mathfrak{p} \implies a \in \mathfrak{p}$ or $b \in \mathfrak{p}$, indicating the existence of irreducible elements in $\mathfrak{p}$. Then, $\mathfrak{p}$ not being principal gives the existence of two coprime elements, otherwise g would be a multiple of f and $\mathfrak{p}$ would be principal.

(b) Applying Bézout's identity yields, for $A, B \in \mathbb{C}(x)[y]$:

$$Af + Bg = 1.$$

Clearing denominators then gives $A'f + B'g = h(x)$ for $A', B' \in \mathbb{C}[x,y]$ and $h \in \mathbb{C}[x]$.

(c) Consider $h(x)$ from before and the intersection $\mathfrak{p} \cap \mathbb{C}[x]$, which is prime in $\mathbb{C}[x]$. This means that $\mathfrak{p} \cap \mathbb{C}[x] = (x - \alpha)$ for some $\alpha \in \mathbb{C}$ and thus $(x - \alpha) \in \mathfrak{p}$. By symmetry, we can conclude the same swapping $x \leftrightarrow y$. Hence, $(x - \alpha, y - \beta) \subseteq \mathfrak{p}$ and, by maximality, $\mathfrak{p} \subseteq (x - \alpha, y - \beta)$, telling us that $\mathfrak{p} = (x - \alpha, y - \beta)$.

Solution to Exercise 2.10

Given that $\mathbb{R}[x]$ is a Euclidean domain, we can use similar techniques to the previous examples to deduce that the primes are (0), $(x - \alpha)$ for $\alpha \in \mathbb{R}$ and $(x^2 + bx + c)$ for $b^2 < 4c$. Take $f = x^3 + x + 1$. Then:

$$f((0)) = f(x),$$

whilst

$$f((x - \alpha)) \equiv \alpha^3 + \alpha + 1$$

merely corresponds to evaluation. Finally, considering $(x^2 + 1)$ we get

$$f((x^2 + 1)) \equiv -x + x + 1 \equiv 1.$$

Solution to Exercise 2.11

We have (0) and $(x - \alpha)$ for each $\alpha \in \mathbb{F}_q$. Moreover, $(f) \in \operatorname{Spec}(\mathbb{F}_q[x])$ if f is irreducible. These can be of any degree. In particular, these points correspond to sets of Galois conjugates in $\overline{\mathbb{F}}_q$. To see this in action, consider $f = x^2 + x + 1$ and $q = 2$. Then, $f = 0$ has no solutions over $\mathbb{F}_2$. Over $\overline{\mathbb{F}}_2$ (specifically some extension isomorphic to $\mathbb{F}_4$), $f = x^2 + x + 1 = (x - \alpha)(x - \alpha^2)$. Given $\operatorname{Gal}(\mathbb{F}_4 \mid \mathbb{F}_2) = \langle \operatorname{Frob}_2 \rangle$, $\operatorname{Frob}_2(\alpha) = \alpha^2$.

The same is essentially true for $\mathbb{A}^1_{\mathbb{Q}}$. The irreducible (degree > 1) polynomials correspond to a single point of an entire Galois conjugacy class of algebraic numbers. For example, take

$\mathfrak{p} = (x^3 - 2)$ (irreducible due to Eisenstein with $p = 2$). Then, the residue field $\mathbb{Q}[x]/(x^3 - 2)$ either sends $x \mapsto \sqrt[3]{2}$, $x \mapsto \zeta_3\sqrt[3]{2}$, or $x \mapsto \zeta_3^2\sqrt[3]{2}$. Another way to see this is with the induced map $\mathbb{A}^1_{\overline{\mathbb{Q}}} \to \mathbb{A}^1_{\mathbb{Q}}$ sending $(x - \alpha) \mapsto (m_\alpha(x))$ (the ideal generated by the minimal polynomial over $\mathbb{Q}$). The map can then be seen by essentially "folding" the Galois orbit of α into just one element $m_\alpha(x)$.

Solution to Exercise 2.12

Suppose that $(f_1), \ldots, (f_n) \in \operatorname{Spec}(K[x])$ is an exhaustive list of the points. Let $N = \prod_i f_i + 1$. Given that $K[x]$ is an integral domain and each f_i has degree at least one, $\deg(N) > 0$. Given $K[x]$ is also a UFD, it must be divisible by an irreducible polynomial q (given our list was exhaustive, this must be one of our f_i elements).

Given that $q \mid N$, it must clearly divide $\prod_i f_i$ and thus the difference $N - \prod_i f_i = 1$. The only elements which divide 1 are the invertible elements ($K[x]^\times = K^\times$), as K is a field. However, q cannot be a degree zero polynomial, thus contradicting the list of points being finite.

Solution to Exercise 2.17

For $\mathfrak{p} \in \operatorname{Spec} B$, if $ab \in \varphi^{-1}(\mathfrak{p})$ then $\varphi(ab) = \varphi(a)\varphi(b) \in \mathfrak{p}$, meaning that $\varphi(a) \in \mathfrak{p}$ or $\varphi(b) \in \mathfrak{p}$. Equivalently, $a \in \varphi^{-1}(\mathfrak{p})$ or $b \in \varphi^{-1}(\mathfrak{p})$, meaning that $\varphi^{-1}(\mathfrak{p}) \in \operatorname{Spec}(A)$.

Solution to Exercise 2.21

a. $I = (b - a^2)$ and $J = (y - x^2, z - y^2)$, and $(x, y, z) \mapsto (a, b, b^2)$.

b. Take $\mathfrak{q} \in \operatorname{Spec}(B/\mathfrak{b})$. Then $\mathfrak{b} \subseteq \mathfrak{q}$ and $\varphi(\mathfrak{a}) \subseteq \mathfrak{b}$ indicates that

$$\mathfrak{a} \subseteq \varphi^{-1}(\mathfrak{b}) \subseteq \varphi^{-1}(\mathfrak{q}).$$

Thus, $\varphi^{-1}(\mathfrak{q}) \in \operatorname{Spec}(A/\mathfrak{a})$.

c. This reduces to computing $\varphi^{-1}(\mathfrak{m}_{a_1,\ldots,a_m}) \in \operatorname{Spec}(K[y_1, \ldots, y_n])$. Suppose $P(y_1, \ldots, y_n) \in \varphi^{-1}(\mathfrak{m}_{a_1,\ldots,a_m})$, then $\varphi(P) = P(f_1(x_1, \ldots, x_m), \ldots, f_n(x_1, \ldots, x_m)) \in \mathfrak{m}_{a_1,\ldots,a_m}$. The polynomials $\varphi(P)$ which are in $\mathfrak{m}_{a_1,\ldots,a_m}$ are those which vanish at $x_i = a_i$ for $1 \leq i \leq m$. Thus, $\varphi^{-1}(\mathfrak{m}_{a_1,\ldots,a_m})$ sends $(a_1, \ldots, a_m)$ to $(f_1(a_1, \ldots, a_m), \ldots, f_n(a_1, \ldots, a_m)) \in K^n$.

Solution to Exercise 2.22

The "order preserving" component comes from the fact that extensions and contractions preserve order. Moreover, contractions preserve primality, so it suffices to show just the other direction.

For $\mathfrak{p} \in \operatorname{Spec} A$, $A/\mathfrak{p}$ is an integral domain. Let $\overline{S}$ be the image of S in $A/\mathfrak{p}$. Then $S^{-1}A/S^{-1}\mathfrak{p} \cong \overline{S}^{-1}(A/\mathfrak{p})$. This tells us that either $\overline{S}^{-1}(A/\mathfrak{p}) = 0$ or $\overline{S}^{-1}(A/\mathfrak{p}) \subseteq \operatorname{Frac}(A/\mathfrak{p})$. Equivalently, $S^{-1}\mathfrak{p}$ is either the unit ideal or prime, where the former happens if $\mathfrak{p}$ meets S.

Solution to Exercise 2.25

We consider the map $\Phi : (\mathbb{C}[x,y]/(xy))_x \to \mathbb{C}[x]_x$ given by

$$\frac{[f(x,y)]}{x^n} \mapsto \frac{f(x,0)}{x^n},$$

where $[f(x,y)]$ denotes the image of the polynomial in the quotient $\mathbb{C}[x,y]/(xy)$.

The map is surjective as any $p(x)/x^n \in \mathbb{C}[x]_x$ is "hit" by $[p(x)]/x^n \in (\mathbb{C}[x,y]/(xy))_x$. Now, if $\Phi([f(x,y)]/x^n) = 0$ then $f(x,0)/x^n = 0$, meaning that $f(x,y)$ has no purely x terms (i.e. $f(x,y) = yg(x,y)$ for some $g(x,y) \in \mathbb{C}[x,y]$. As x is a unit in $(\mathbb{C}[x,y]/(xy))_x$, we see that

$$\frac{[f(x,y)]}{x^n} = \frac{[yg(x,y)]}{x^n} = \frac{[xy][g(x,y)]}{x^{n+1}} = 0,$$

given the relation $xy = 0$. Thus, the kernel is trivial, and hence we have an isomorphism of the above rings.

Solution to Exercise 2.28

$(0) \notin \operatorname{Spec}(K[\varepsilon]/(\varepsilon^2))$ as it does not contain (x^2). Let $f \in K[x]$ be irreducible, then $(x^2) \subseteq (f)$ means that $f \mid x^2$. Thus, $f = kx$ for $k \in K$, meaning that we are only left with (x). Thus, $\operatorname{Spec} K[\varepsilon]/(\varepsilon^2) = \{(\varepsilon)\}$.

Solution to Exercise 2.31

Suppose d is a quadratic residue. Then, $\pi^{-1}((p)) = \operatorname{Spec} \mathbb{F}_p[x]/(x^2-d) = \operatorname{Spec}(\mathbb{F}_p[x]/(x-\alpha) \times \mathbb{F}_p[x]/(x-\overline{\alpha})) = \operatorname{Spec}(\mathbb{F}_p \times \mathbb{F}_p)$, giving us that p splits.

Suppose now that d is not a quadratic residue. Then, $(x^2 - d)$ is irreducible mod p and $\mathbb{F}_p[x]/(x^2-d)$ is a field, meaning that p is inert.

Solution to Exercise 2.32

a. Take $\overline{\mathfrak{p}} \in \operatorname{Spec}(A/\mathfrak{n})$. Then, in A, $\pi^{-1}(\overline{\mathfrak{p}})$ corresponds to a prime containing $\mathfrak{n}$: $\mathfrak{n} \subseteq \pi^{-1}(\overline{\mathfrak{p}})$. However, this is always true in the case of ideals of nilpotents, as if $n \in \mathfrak{n}$ then $n^k = 0$ for some $k > 1$. Thus, $n^k = 0 \in \mathfrak{p}$ and inductively we get $n \in \mathfrak{p}$. Hence, the inclusion is a bijection of sets.

b. Check closure under addition and by multiplication from elements in A. It is easy to see why the nilradical is contained within $\cap_{\mathfrak{p} \in \operatorname{Spec}(A)} \mathfrak{p}$, the reverse inclusion requires supposing that $a \notin \mathfrak{N}(A)$ and applying Zorn's lemma to prove that there is a prime ideal not containing a.

Solution to Exercise 3.1

a. Take $S = \varnothing, A$.

b. The forward inclusion is straightforward. The reverse inclusion is obtained as follows. Let $\mathfrak{p} \in V(\sum_i \mathfrak{a}_i)$. Then, $\sum_i \mathfrak{a}_i \subset \mathfrak{p}$. Then, $\mathfrak{a}_j \subseteq \mathfrak{p}$ for all j. Given that $\mathfrak{p}$ contains every ideal $\mathfrak{a}_j$, it belongs to every vanishing set. In other words, $\mathfrak{p} \in \cap_i V(\mathfrak{a}_i)$.

c. Forward inclusion here is straightforward. If $\mathfrak{p} \in V(\mathfrak{a}_1\mathfrak{a}_2)$ then $\mathfrak{a}_1\mathfrak{a}_2 \subset \mathfrak{p}$, meaning that either $\mathfrak{a}_1 \subset \mathfrak{p}$ or $\mathfrak{a}_2 \subset \mathfrak{p}$, exactly what is required.

Solution to Exercise 3.5

$$\begin{aligned}\pi^{-1}(V(\mathfrak{a})) &= \{\mathfrak{p} \in \operatorname{Spec} A : \mathfrak{a} \subseteq \varphi^{-1}(\mathfrak{p})\} \\ &= \{\mathfrak{p} \in \operatorname{Spec} A : (\varphi(\mathfrak{a})) \subseteq \mathfrak{p}\} = V((\varphi(\mathfrak{a}))).\end{aligned}$$

Solution to Exercise 3.6

a. $\operatorname{Spec}(B/\mathfrak{b}) = V(\mathfrak{b}) \underset{\text{closed}}{\subseteq} \operatorname{Spec}(B)$. On the other hand, we saw that B_f was the set where f did not vanish, meaning that $\operatorname{Spec}(B_f) = \operatorname{Spec}(B) \setminus V((f))$.

b. Take $B = \mathbb{Z}$ and $S = \mathbb{Z}^\times$. Then, $\operatorname{Spec}(\mathbb{Q}) = \{(0)\} \subseteq \operatorname{Spec}(\mathbb{Z})$. Clearly it is not closed ((0) is the generic point of $\operatorname{Spec}\mathbb{Z}$). Possibly the definition of the distinguished open set can show that this is not open either.

References

[Belmans, 2011] Belmans, P. (2011). Atlas of spec($\mathbb{Z}[x]$).

[McLarty, 2016] McLarty, C. (2016). How grothendieck simplified algebraic geometry. *Notices of the AMS*, 63(3):256–265.

[Vakil, 2025] Vakil, R. (2025). The rising sea: Foundations of algebraic geometry.